

Curriculum Vitae

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aabbtree77.github.io

1 Education

1995-1999: BSc (EE), Vilnius Gediminas Technical University.
1999-2008: MSc (Tech), DSc (Tech), Helsinki University of Technology.

2 Programming Experience

2.1 Web

Tools: VPS, ssh, Linux, git, Porkbun, Caddy, Go, React, Postgres, SQLite, sqlc, Makefile, Docker Compose, Tailwind, Excalidraw, Mermaid. Also tried: MongoDB Atlas, render.com, MaxMind Geolocation, MERN, Drizzle ORM, Astro.

2.1.1 A Message Board

github.com/aabbtree77/opt

A public bulletin board with proof of work and rate limiters. Build, debug, dev, prod environments. It ran as *initials.dev* on Hetzner (CAX11) for about three months.

2.1.2 Other Projects

- github.com/aabbtree77/schatzhauser
- github.com/aabbtree77/miniguestlog
- github.com/aabbtree77/site-redesign-demo

2.2 Embedded

Tools: ATmega, avr-gcc, ESP32, Ubuntu, routers, MQTT, I2C, libp2p. Wrote firmware for industrial machines (poster printers, box cutters, gym equipment), ensured global connectivity.

2.2.1 Adast Maxima MS80

github.com/aabbtree77/adast

Wrote firmware (C, avr-gcc) for the paper cutting machine, ensuring correctness and precision under heavy factory workloads, with some updates throughout the years. The machine is still in use today (2026).

2.2.2 Other Projects

- github.com/aabbtree77/esp32-vpn
- github.com/aabbtree77/atmega8-ssd1306-ds18b20-dht22
- github.com/aabbtree77/digispark-attiny85-command-line

2.3 3D

Tools: OpenGL, GLSL, C++, Nim, Go, Blender, GLFW, GLTF, GTX 760. OpenFOAM, ParaView, ProActive PACA Grid (INRIA). Shape analysis (MDS, geodesics).

2.3.1 Volumetric Lighting

github.com/aabbtree77/twinpeekz

A complete rendering pipeline with one of the most beautiful special effects - volumetric lighting.

2.3.2 Automotive Optimization

hal.archives-ouvertes.fr/hal-00723427

A large-scale engineering optimization project involving Navier–Stokes simulations, grid computing, meta-optimization, kriging, Bayesian methods, and Monte Carlo analysis, executed end-to-end on cloud infrastructure in collaboration with academic and industrial partners.

2.3.3 Notable Bug Fixes

- github.com/go-gl/gl/issues/128
- github.com/doitsujin/dxvk/issues/3789
- github.com/Max1412/Graphics2/issues/24
- github.com/godlikepanos/anki-3d-engine/issues/59

2.4 Data Analysis

Tools: Matlab, Scilab, Python. Kriging, Bayes, CMA-ES.

2.4.1 Example Project: MNIST Digits

github.com/aabbtree77/MNIST-0.17

This repo contains two world records in the MNIST digit recognition competition. One is by me within classics (log-Gabors with max pooling and kriging) - 0.29%, and the other is a confirmed absolute world record with deep learning by Jonas Matuzas - 0.17%.

2.4.2 In Search of the Best Derivative-Free Optimization Algorithm

github.com/aabbtree77/discombobulated

A lot of tests and some insights into ES, CMAES, ARRDE algorithms with BBOB-2009 and CEC-2017. Links to discussions with the authors, links to some issues with even more testing. It seems the ARRDE is the winner.

2.4.3 Other Projects

- github.com/aabbtree77/imdb-sqlite-queries
- github.com/aabbtree77/uci-marketing-analysis-cart
- aabbtree77.github.io/tensors
- github.com/polydojo/jispy/issues/1

3 Employment

01.02.2014-Now	Self-Funded.
01.02.2013-31.01.2014	Faculty of Informatics at the University of Lugano, Via Giuseppe Buffi 13, CH-6904 Lugano, Switzerland.
01.09.2011-31.12.2012	DEMO, École Nationale Supérieure des Mines de Saint-Étienne, 158 cours Fauriel, 42023 Saint-Étienne, CEDEX2, France.
01.09.2008-31.08.2009	Department of Neurobiology at the University of California, Los Angeles, 73-235 CHS, Los Angeles, CA 90095-1763.
01.05.2000-31.07.2008	Faculty of Information and Natural Sciences at Helsinki University of Technology, P.O.Box 5400 (Konemiehentie 2), FI-02015, Finland.

4 Publications

1. D. Boscaini, R. Girdziušas, and M. M. Bronstein (2014). "Coulomb shapes: using electrostatic forces for deformation-invariant shape representation." *Eurographics W. 3D Object Retrieval*. diglib.org/handle/10.2312/3dor.20141044.009-015.
2. R. Girdziušas, J. Janusevskis, and R. Le Riche (2012). "On integration of multi-point improvements." *NIPS W. Bayesian Opt. Dec. Making*. www.emse.fr/~leriche/integ_multiEI_nips2012_final.pdf.
3. R. Le Riche, R. Girdziušas, and J. Janusevskis (2012). "A Study of Asynchronous Budgeted Optimization." *NIPS W. Bayesian Opt. Dec. Making*. www.emse.fr/~leriche/synchro_asynchro_final.pdf.
4. R. Girdziušas, R. Le Riche, et al. (2012). "Parallel Budgeted Optimization Applied to the Design of an Air Duct." hal.archives-ouvertes.fr/hal-00723427.
5. R. Le Riche, R. Girdziušas, D. Villanueva, et al. (2012). "Squaring the engineering optimization circle: Distributed global optimization algorithms for computationally expensive problems." www.emse.fr/~leriche/rio2012_LeRiche.pdf.

6. J. Janusevskis et al. (2012). "Expected Improvements for the Asynchronous Parallel Global Optimization of Expensive Functions: Potentials and Challenges." *LNCS*. vol. 7219, pp. 413–418.
7. R. Girdziušas and D. Ringach (2009). "The Jacobian Singularity in the Thomson Problem. Unpublished." aabbtree77.github.io/pdfs/ucla2009.pdf.
8. R. Girdziušas (2008). "Stability and Inference in Discrete Diffusion Scale-Spaces. DSc (PhD) thesis, 2008." aaltodoc.aalto.fi/handle/123456789/2999. Advisors: E. Oja and J. Laaksonen (Helsinki Univ. of Tech.). External examiners: S. Siltanen (Tampere Univ. of Tech.) and K. Ruotsalainen (Univ. of Oulu). Opponents: S. Siltanen and A. Gotchev (Tampere Univ. of Tech.).
9. R. Girdziušas and J. Laaksonen (2007b). "When Is a Discrete Diffusion a Scale-Space?" *ICCV*. dx.doi.org/10.1109/ICCV.2007.4408895.
10. R. Girdziušas and J. Laaksonen (2007a). "How Marginal Likelihood Inference Unifies Entropy, Correlation and SNR-based Stopping in Nonlinear Diffusion Scale-Spaces." *LNCS*. vol. 4843, pp. 811–820.
11. R. Girdziušas and J. Laaksonen (2005b). "Jacobi Alternative to Bayesian Evidence Maximization." *LNCS*. vol. 3697, pp. 247–252.
12. R. Girdziušas and J. Laaksonen (2005e). "Use of Input Deformations with Brownian Motion." *LNCS*. vol. 3686, pp. 219–228.
13. R. Girdziušas and J. Laaksonen (2005d). "Optimal Stopping and Constraints for Diffusion Models of Signals with Discontinuities." *LNAI*. vol. 3720, pp. 576–583.
14. R. Girdziušas and J. Laaksonen (2005c). "Optimal Ratio of Lamé Moduli with Application to Motion of Jupiter Storms." *LNCS*. vol. 3540, pp. 1096–1106.
15. R. Girdziušas and J. Laaksonen (2005a). "Gaussian Processes of Nonlinear Diffusion Filtering." *IJCNN*, pp. 1012–1017. dx.doi.org/10.1109/IJCNN.2005.1555991.
16. R. Girdziušas and J. Laaksonen (2004). "Gaussian Process Regression with Fluid Hyperpriors." *LNCS*. vol. 3316, pp. 567–572.
17. M. Aksela et al. (2003). "Methods for Adaptive Combination of Classifiers." *IJDAR* vol. 6.1, pp. 23–41.
18. M. Aksela et al. (2002). "Class-Confidence Critic Combining." *IWFHR*, pp. 201–206. dx.doi.org/10.1109/IWFHR.2002.1030909.
19. R. Girdziušas (2001). "Discriminative Online Recognition of Isolated Handwritten Characters. MSc thesis. Helsinki Univ. of Tech."
20. R. Girdziušas and J. Laaksonen (2001). "Multilayer Perceptron Approach to Non-Rigid Image Matching." *ICONIP*. vol. 1, pp. 491–496.

5 Talks

- Presentations: ICONIP 2001, 2004; ICAPR 2005; ICANN 2005; OMD2 meetings (Paris 2011, 2012).
- Posters: SCIA 2005; IJCNN 2005; ECML 2005; ICCV 2007; ACCV 2007.

- Invited talk at the DIKU group, the University of Copenhagen, 2008.

6 Awards

01.09.1999- 01.06.2000	Nokia scholarship (Natalia Kantola, Robertas Tamulevičius).
01.01.2002- 01.07.2006	ComMIT grant (postgraduate studies).
09.11.2005- 09.15.2005	ICANN travel grant.
07.01.2006- 12.01.2006	HUT scholarship.
01.22.2008- 01.24.2008	Travel grant (Prof. Mads Nielsen).
04.03.2008:	Doctoral thesis award: "Pass with Distinction".

7 Recommendation Letters

- Prof. Erkki Oja, Konemiehentie 2, P.O. Box 5400, 02015 HUT, Finland.
Email: Erkki.Oja@hut.fi, phone: +358-9-451 3265, fax: +358 9 451 3277.
- Doc. Jorma Laaksonen, Konemiehentie 2, P.O. Box 5400, 02015 HUT, Finland.
Email: Jorma.Laaksonen@hut.fi, phone: +358-9-451 3269, fax: +358 9 451 3277.
- Prof. Dario Ringach, Dep. of Neurobiology, UCLA, Los Angeles, CA 90095-1563.
Email: dario@ucla.edu, phone: +310 825 6606.
- Doc. Rodolphe Le Riche, DEMO, 158 cours Fauriel, 42023 Saint-Étienne, CEDEX 2.
Email: leriche@emse.fr, phone: +336 31 262 678.
- Prof. Michael M. Bronstein, SI-109, Via Giuseppe Buffi 13, 6904, Lugano, Switzerland.
Email: michael.bronstein@usi.ch, phone: +41 58 666 4120.
- Saulius Rakauskas (senior electronics engineer at Infovega UAB, retired). Antakalnio 48-16, LT-10304, Vilnius, Lietuva.
Email: radiomtx@gmail.com, phone: +370 652 42317.
- Bronius Grigas (the founder of Mituvėlė UAB). Rodūnios kl. 11, LT-02189, Vilniaus m. sav., Lietuva.
Email: bronius@mituvele.lt, phone: +370 686 43642.

8 Additional Information

Date of Birth: 30.12.1977.
Citizenship: Lithuanian (Lithuania, European Union).
Languages: English (7/10), Russian (8/10), Lithuanian (9/10).
Driver's Licence: B.
Sports: 2318 (hyperboloid777 @ lichess), Kyokushin 2nd belt (1993).
Erdős Number: 5 (Bollobas, Mannila, Bingham, Oja), 4 (Odlyzko, Guibas, Bronstein).
Hobbies: Classical and modern cinema.